

Tutorial 4: Discrete Probability Distribution (STUDENT'S VERSION)

Theme: Statistical Functions and Distributions in R

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

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1 A. More on Statistical Functions in R

Key Definition / Result

Core Utility and Summary Functions. Students must become comfortable with R commands that describe samples and generate random data.

- **Median:** `median(c(1,4,2,5,8))` → Outputs 4
- **Quantile:** `quantile(c(1,4,2,5,8), 0.75)` → Outputs 5
- **Random Sampling:** `sample(1:5, 3, replace=T)` → Outputs e.g., 1 4 4 (Random!)
- **Histogram:** `hist(c(1,4,2,5,8))` → Displays sample distribution shape
- **Empirical CDF:** `plot(ecdf(runif(100)))` → Displays Empirical Cumulative Distribution Function plot

Suggested AI Prompt: Descriptive Statistics and Interpretation ★ Core

Show me how to use R to calculate descriptive statistics such as mean, median, quantile, standard deviation, etc. for a given sample of data. Explain what these statistics measure and how they are useful for risk management.

Suggested AI Prompt: Generating Random Scenarios

★ Core

Show me how to use R to generate random numbers by the sample function.

Explain what each argument in the sample function means and how it affects the distribution.

2 B. Working with Distribution Functions

Key Definition / Result

The Four Function Prefixes in R. Probability distributions in R follow a strict naming convention. You combine a prefix with the distribution's short name (e.g., `unif`, `binom`):

- **d** – density / probability mass function (pdf/pmf). Finds the probability of a specific value.
- **p** – cumulative distribution function (cdf). Finds the probability less than or equal to a value.
- **q** – quantile function (inverse cdf). Finds the value given a probability.
- **r** – random number generation. Draws random samples from the distribution.

Core Distributions and their Arguments:

- **Uniform:** `unif` (Requires: `min`, `max`)
- **Binomial:** `binom` (Requires: `size`, `prob`)

Suggested AI Prompt: Executing the R Syntax

★★ Insight

Explain what distribution functions are in R and how to use them. Show me some examples of how to use `d`, `r`, `p`, and `q` functions for the Uniform, Bernoulli, and Binomial distributions in risk management scenarios.

3 C. The Bernoulli and Binomial Distributions

Key Definition / Result

The Bernoulli Distribution is the simplest discrete distribution, representing a single trial with only two possible outcomes (e.g., Success/Failure, Default/No Default). It requires one parameter: p (probability of success).

The Binomial Distribution represents the number of successes in n independent Bernoulli trials. In R, we do not have a standalone "Bernoulli" function. Instead, we use the Binomial functions and set the `size` argument to 1.

Worked Example: Simulating a Bernoulli Trial

★ Core

Question. Simulate 5 individual loan evaluations where the probability of default (success) is 15%.

Worked Example: Calculating Binomial Probabilities

★ Core

Question. A portfolio has 10 independent loans. Each has a 15% chance of default. Find the probability that exactly 3 loans default, and the probability that 3 or fewer default.

Investigation / Activity: Visualising the Binomial PMF**★★ Insight**

Task. Plot the Probability Mass Function (PMF) of a Binomial distribution representing the number of claims from 20 policies, where each policy has a 30% chance of claiming. What happens to the shape if you change the probability to 0.5?

4 D. The Uniform Distribution (Discrete & Continuous)

Investigation / Activity: Discrete vs Continuous Uniform**★★ Insight**

Task. The `runif` function generates *continuous* uniform variables. How can we simulate rolling a fair 6-sided die (a *Discrete Uniform* distribution)?

5 E. Comparing Empirical and Theoretical Distributions

Investigation / Activity: ECDF vs. Theoretical CDF (Uniform)

★★

Insight

Tasks for the student:

1. Generate a sample of random numbers from a Uniform Distribution (e.g., $X = \text{runif}(100, 0, 100)$).
2. Draw the ECDF for the sample, and draw the theoretical CDF for the uniform distribution.
3. From the ECDF plot, find the 0.2 quantile of the sample. Compare with the results of `quantile(X, 0.2)` and `qunif(0.2)`.
4. Repeat the above analysis using a larger sample size. What do you observe?

6 F. Exercise: Convergence of Quantiles

Investigation / Activity: Sample vs. Population Quantiles

★★★

Advanced

Task. For each of the `unif` and `binom` distributions, generate 1,000 samples, then compute the 0.95 and 0.99 quantiles. Are they close to the population quantiles? Repeat the analysis using 100,000 samples instead of 1,000. Discuss your findings below.

Copyright, Compliance, and Attribution

Key Definition / Result

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